



# TESS Investigation—Demographics of Young Exoplanets (TI-DYE). III. An Inner Super-Earth in TOI2076

Madysen G. Barber<sup>1,3</sup> , Andrew W. Mann<sup>1</sup> , Andrew Vanderburg<sup>2</sup> , Andrew W. Boyle<sup>1,3</sup> , and Ana Isabel Lopez Murillo<sup>1,4</sup>

<sup>1</sup> Department of Physics and Astronomy, The University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA; [madysonb@live.unc.edu](mailto:madysonb@live.unc.edu)

<sup>2</sup> Department of Physics and Kavli Institute for Astrophysics and Space Research, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

Received 2025 April 24; revised 2025 May 5; accepted 2025 May 8; published 2025 June 12

## Abstract

Young (<500 Myr) multiplanet transiting systems are valuable environments for understanding planet evolution by offering an opportunity to make direct comparisons between planets from the same formation conditions. TOI 2076 is known to harbor three  $2.5\text{--}4 R_{\oplus}$  planets on 10–35 days orbits. All three are JWST cycle 3 targets (for transmission spectroscopy). Here, we present the detection of TOI 2076 e; a smaller ( $1.35 R_{\oplus}$ ) inner (3.02 days) planet in the system. We update the age of the system by analyzing the rotation periods, lithium equivalent widths, color–magnitude diagram, and variability of likely comoving stars, finding that TOI 2076 and comoving planetary system TOI 1807 are  $210 \pm 20$  Myr. The discovery of TOI 2076 e is motivation to revisit known transiting systems in search of additional planets that are now detectable with new TESS data and updated search methods.

*Unified Astronomy Thesaurus concepts:* Exoplanets (498); Transit photometry (1709); Exoplanet evolution (491); Young star clusters (1833)

*Materials only available in the online version of record: data behind figure, machine-readable table*

## 1. Introduction

Population level analyses of Kepler and K2 transiting systems show that mature multiplanet systems are largely tightly packed (P. S. Muirhead et al. 2015; J. M. Brewer et al. 2018), having a high level of intrasystem uniformity in radius and period space that is much more similar than expected by chance (J. J. Lissauer et al. 2011; R. I. Dawson et al. 2016; L. M. Weiss et al. 2018). The current population of known young multiplanet systems do not show this same level of intrasystem uniformity (P. C. Thao et al. 2024), seemingly suggesting that planet systems become uniform over time.

However, the mature Kepler sample is dominated by small planets ( $\lesssim 4 R_{\oplus}$ ; E. A. Petigura et al. 2022), while young planets in TESS are overwhelmingly large ( $\gtrsim 4 R_{\oplus}$ ; R. B. Fernandes et al. 2022; S. Vach et al. 2024). In order to determine if intrasystem uniformity is primordial or an evolutionary effect, we need to be consistently sensitive to the smallest planets in the youngest systems.

One challenge of detecting small planets around young stars is the high levels of stellar activity present. This obscures small planet signals and can adversely affect light curve processing. However, custom light curve processing (e.g., A. Vanderburg et al. 2019) and search methods tuned for young stars (e.g., A. C. Rizzuto et al. 2017) have enabled the discovery of younger (e.g., M. G. Barber et al. 2024a) and smaller (e.g., A. W. Mann et al. 2018) planets. HD 63433 d, an Earth-sized planet on a tight orbit around a member of the 400 Myr Ursa Major group (B. K. Capistrant et al. 2024), demonstrates how

processing instrumental systematics and stellar variability simultaneously could increase our sensitivity to additional, smaller planets in known young transiting systems. This system also highlights the need for such work to study the origins of intrasystem uniformity; the original two planets had similar radii, but the additional smaller planet was a mismatch in both radius and period (low intrasystem uniformity).

As part of our larger survey searching for planets in young stellar associations (the TI-DYE survey; M. G. Barber et al. 2024a, 2024b), we revisit known transiting systems searching for missed planets. With our increased sensitivity, we are able to recover smaller planets that could aid in understanding whether uniformity results from formative (e.g., S. T. S. Poon et al. 2020) or evolutionary processes (e.g., E. D. Lopez & J. J. Fortney 2013).

Young multiplanet systems are also found near period resonances at a much higher rate than their older counterparts (F. Dai et al. 2024). In turn, they are more likely to exhibit detectable transit-timing variations (TTVs; A. I. Lopez Murillo 2025, in preparation). Given the challenges of measuring radial velocities in the presence of stellar noise (S. Blunt et al. 2023), TTVs remain a more promising route for measuring masses and eccentricities of young planets. However, this method only works if we detect all the major perturbing planets.

Since the discovery, TOI 2076 has been observed in two additional TESS sectors, and we have made numerous improvements to our transit-search pipeline (see M. G. Barber et al. 2024b), motivating a new search. Here, we report the discovery of an inner, smaller, fourth planet in the system: TOI 2076 e. Due to different prior analyses yielding different ages for TOI 2076, we rederive the age using the rotations, lithium levels, color–magnitude diagram positions, and variabilities of identified comoving stars. Given the systems prior TTV detection (H. P. Osborn et al. 2022; A. I. Lopez Murillo et al. 2025, in preparation) and JWST data (A. Feinstein et al. 2024),

<sup>3</sup> NSF Graduate Research Fellow.

<sup>4</sup> UNC Chancellor’s Science Scholar.



Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](https://creativecommons.org/licenses/by/4.0/). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

the discovery of TOI 2076 e and updated age analysis will aid in the interpretation of existing data on the system.

In this paper, we detail the discovery and age analysis of TOI 2076 e. We describe our light curve extraction in Section 2 and our transit search in Section 3. Taking into account the stellar parameters described in Section 4, we derive the planet parameters in Section 5, rule out false-positive scenarios in Section 6, and run an injection-recovery analysis in Section 7. We search for comoving stars to TOI 2076 in Section 8 and use these members to determine an age of the cluster in Section 9. We discuss potential additional known planets in the cluster in Section 10 and search for additional transiting systems in Section 11. Finally, we discuss the impact of this new planet and age analysis in Section 12.

## 2. TESS Light Curve

TOI 2076 (TIC 27491137) was first observed by TESS in Sector 16, from 2019 September 12 to 2019 October 6, and reobserved in Sector 23 (2020 March 19–2020 April 15), Sector 50 (2022 March 26–2022 April 22), and Sector 77 (2024 March 25–April 23). The target was preselected for 2 minute short-cadence light curves for Sectors 16 and 23 and 20 second short-cadence light curves for Sectors 50 and 77. During Sector 77, TESS was put into Safe Mode April 6–16 due to the onboard data recorder filling up due to a missing datalink on April 1. Thus, Sector 77 contains only 17.92 days of science data.<sup>5</sup> All TESS light curves used in this analysis can be found in MAST: doi: [10.17909/t9-st5g-3177](https://archive.stsci.edu/missions/teess/doc/teess_drm/teess_sector_77_drm108_v01.pdf) and [10.17909/t9-nmc8-f686](https://archive.stsci.edu/missions/teess/doc/teess_drm/teess_sector_77_drm108_v01.pdf).

### 2.1. Extraction Pipeline

For our analysis, we used a custom light curve extraction pipeline starting from the Simple Aperture Photometry (SAP; J. D. Twicken et al. 2010; R. L. Morris et al. 2020) from the Science Processing Operations Center (SPOC; J. M. Jenkins et al. 2016), utilizing the shortest cadence available for each sector. Our systematic corrections generally followed A. Vanderburg et al. (2019). To summarize, we corrected the SPOC SAP light curves using a linear model consisting of a basis spline with 0.2 day breaks to model low-frequency variations, several moments of the distributions of the spacecraft quaternion time-series measurements within each exposure, seven cotrending basis vectors from the SPOC Presearch Data Conditioning (PDC; J. C. Smith et al. 2012; M. C. Stumpe et al. 2012, 2014) band 3 flux time-series correction with the largest eigenvalues, and a 0.1 day high-pass time series from the SPOC background aperture.

We estimated the uncertainties on the flux by taking the median of three values: (1) the median absolute deviation of the point minus the adjacent point; (2) the median absolute deviation of the flattened light curve (flattened using `lightkurve`; Lightkurve Collaboration et al. 2018, with a  $4\sigma$  outlier threshold); and (3) sigma clipping, applying a median filter, sigma clipping again, and fitting a Gaussian to the resulting distribution of points. Each method was broadly consistent with each other, and the final results are not dependent on the uncertainty estimate. Estimating the uncertainties for each sector separately, and we adopted an uncertainty of 0.0005 for sectors 19 and 23 and 0.001 for sectors 50 and 77.

## 3. Transit Search and Identification of the Planet

Our ongoing search for transiting planets in young stellar associations (M. G. Barber et al. 2024a, 2024b) includes a search for additional transiting planets in known young systems. This program takes advantage of better light curve processing (Section 2.1) and improvements to the transit-search (below). TOI 2076 was one of the first targets where we identified additional transit-like signals.

We used the updated `Notch` and `LOCoR` (N&L; A. C. Rizzuto et al. 2017) pipeline as described in M. G. Barber et al. (2024a). Using a 0.75 day filtering window, `Notch` detrended the light curve using a second-order polynomial while preserving trapezoidal, transit-like signals. At each point, `notch` calculates the change in the Bayesian Information Criterion (BIC) based on the improvement when adding the trapezoid to the model compared to the plain polynomial. We then searched the BIC time-series for periodic signals between 0.5 and 50 days with an SNR  $> 8$  using a box-least-squares (BLS) search. We recovered a 10.4 days signal (TOI 2076 b) with a BLS SNR of 33, a 21.0 days signal (TOI 2076 c) with a BLS SNR of 32, a 35.1 days signal (TOI 2076 d) with a BLS SNR of 56, and a 3.0 day signal (TOI 2076 e) with a BLS SNR of 17.

In Figure 1, we highlight the sensitivity of the combination of the `Notch` pipeline with the custom light curve extraction; the planet was not recovered at sufficient SNR (generally  $> 7$ ) in the PDCSAP or SAP light curve and recovered only at modest SNR ( $\approx 10$ ) using the custom extraction with a more traditional search method (a high-pass filter and BLS).

## 4. Stellar Properties

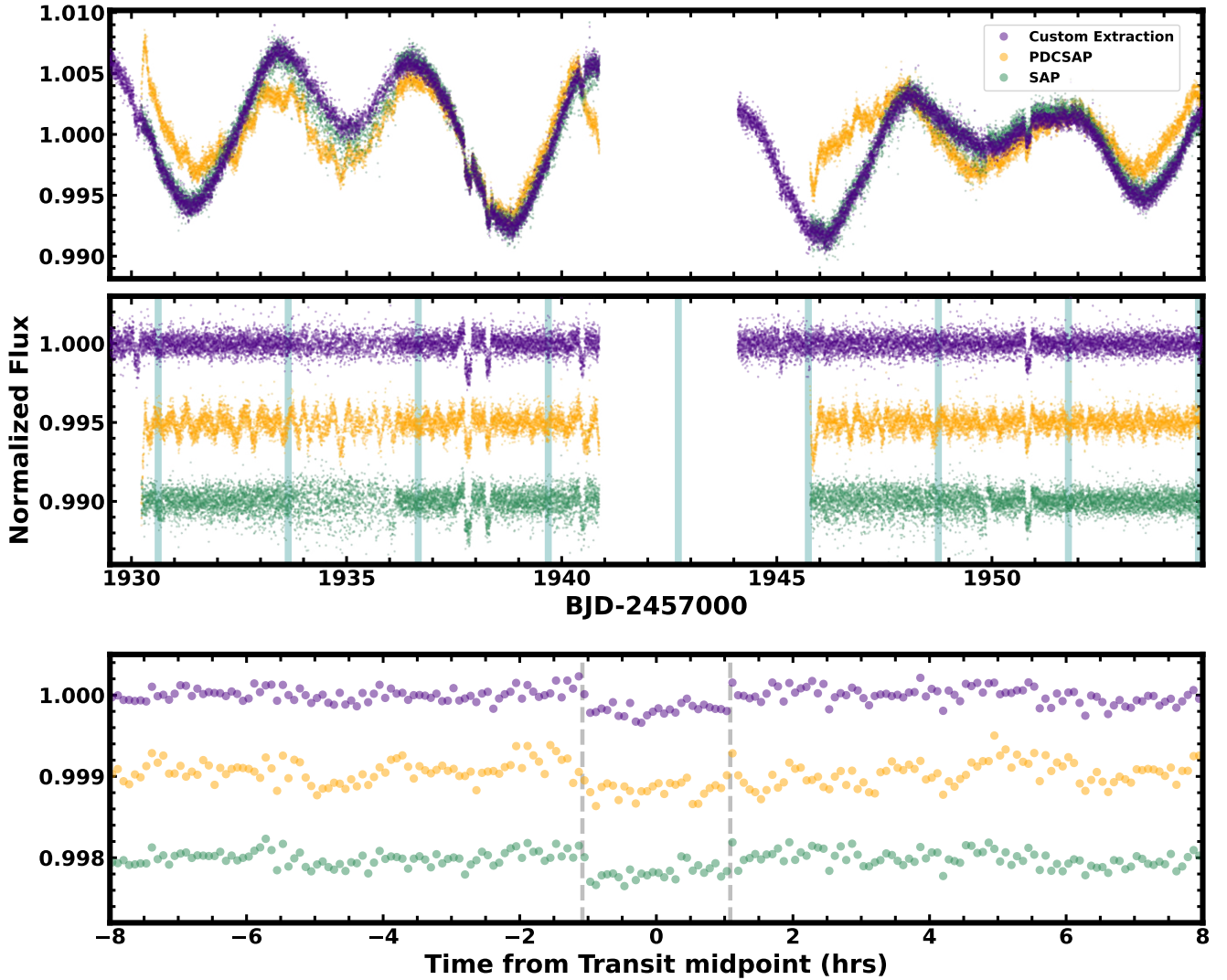
M. G. MacDougall et al. (2023) combined photometry, high-resolution spectroscopy, and Gaia parallaxes to derive stellar properties of 108 TESS planet-hosting stars, including TOI 2076. We chose to adopt their stellar radius ( $R_* = 0.7960^{+0.0286}_{-0.0217}$ ), stellar mass ( $M_* = 0.8557^{+0.0201}_{-0.0212}$ ), effective temperature ( $T_{\text{eff}} = 5192^{+61}_{-58}$ ), and metallicity ( $[\text{Fe}/\text{H}] = 0.020^{+0.058}_{-0.059}$ ) for our transit analysis, though these values are consistent with the stellar values derived in C. Hedges et al. (2021), H. P. Osborn et al. (2022), and M. Damasso et al. (2024).

The four prior studies all estimated the age of TOI 2076:  $204 \pm 50$  Myr from C. Hedges et al. (2021),  $340 \pm 80$  Myr from H. P. Osborn et al. (2022),  $2.7^{+3.6}_{-1.9}$  Gyr from M. G. MacDougall et al. (2023) and  $300 \pm 80$  Myr from M. Damasso et al. (2024; based on the analysis in D. Nardiello et al. 2022). The M. G. MacDougall et al. (2023) age is based on isochronal fitting of just the host star, which tends to give such large uncertainties. The other three ages are all based on comoving stars. While all ages agree, the resulting age uncertainties are larger than what is routinely achieved for young associations (e.g., E. R. Newton et al. 2022; M. L. Wood et al. 2023a; P. C. Thao et al. 2024). Further, inspection of the membership lists shows significant disagreement. Given the importance of this target, we opted to revisit the membership and age of the group in Section 9.

## 5. Transit Analysis and Planet Parameters

We fit the systematics-corrected TESS light curve using MISTTBORN (Markov chain Monte Carlo (MCMC) Interface for Synthesis of Transits, Tomography, Binaries, and Others of

<sup>5</sup> [https://archive.stsci.edu/missions/teess/doc/teess\\_drm/teess\\_sector\\_77\\_drm108\\_v01.pdf](https://archive.stsci.edu/missions/teess/doc/teess_drm/teess_sector_77_drm108_v01.pdf)



**Figure 1.** Our custom-extracted light curve (purple) compared to the PDCSAP (orange) and SAP (green) light curves for Sector 23. The top panel shows the raw, normalized light curves, and the middle panel shows the notch-detrended light curves, using a 0.75 day filtering window, with the transits of TOI 2076 e highlighted in teal and a y offset between the extractions for clarity. The raw and detrended custom light curves show significantly less noise than the PDCSAP light curves. The bottom panel shows the phase-folded light curves binned to 5 minute intervals with a y offset for clarity. The gray dashed lines mark the expected ingress and egress. The custom-extraction light curve shows a clear transit-like shape while the PDCSAP light curve is too noisy to detect the transit. The custom extraction light curve is available in machine readable format as the data behind the figure.

(The data used to create this figure are available in the [online article](#).)

a Relevant Nature; A. W. Mann et al. 2016; M. C. Johnson et al. 2018).<sup>6</sup> MISTTBORN utilizes BATMAN to generate transit models (L. Kreidberg 2015), celerite2 to model the stellar variability using a Gaussian process (GP; D. Foreman-Mackey 2018), and emcee to explore parameter space (D. Foreman-Mackey et al. 2013).

Stochastically-driven damped simple harmonic oscillators (SHOs) have been shown to be preferred in modeling stellar variability (D. Foreman-Mackey et al. 2017) and have been used extensively to model photometry of young stars with multiple transiting planets (e.g., J. M. Wittrock et al. 2022; M. L. Wood et al. 2023a; P. C. Thao et al. 2024). We initially used a sum of two SHOs using the RotationTerm in celerite2. We found that the difference between the quality factors of the two oscillators was unconstrained, suggesting only a single SHO is needed. We instead fit the

stellar variability using a single SHO following the description in E. A. Gilbert et al. (2022). We find that the planetary parameters agree between the single and summed SHOs fits, but the stellar variability is better modeled by the single SHO, and we chose this model as our preferred fit.

We fit for 10 parameters to model the stellar variability and transits of TOI 2076 e simultaneously. For the planet, we fit for the time of inferior conjunction ( $T_0$ ), planet orbital period ( $P$ ), planet-to-star radius ratio ( $R_p/R_*$ ), and impact parameter ( $b$ ). Previous fits of the system suggest the outer three planets are in near circular orbits (A. S. Polanski et al. 2024), so we restrict eccentricity to 0. We also fit for stellar density ( $\rho_*$ ) and two quadratic limb-darkening coefficients ( $q_1$  and  $q_2$ ) following the triangular sampling prescription (D. M. Kipping 2013).

The remaining three parameters were used to describe the GP stellar variability model. We fit for the undamped period of the oscillator ( $P$ ), the damping timescale ( $\tau$ ), and the standard deviation of the process ( $\sigma$ ).

<sup>6</sup> <https://github.com/captain-exoplanet/misttborn>



**Table 1**  
Prior Distributions Used in the MISTTBORN TESS Transit Fit

| Description                 | Parameter                 | Prior <sup>a</sup> |
|-----------------------------|---------------------------|--------------------|
| Planet Parameters           |                           |                    |
| Impact parameter            | $b$                       | $U(0.0, 1.0)$      |
| Planet-to-star radius ratio | $R_P/R_*$                 | $U(0.0, 1.0)$      |
| Stellar Parameters          |                           |                    |
| Limb-darkening coefficient  | $q_1$                     | $U(0.0, 1.0)$      |
| Limb-darkening coefficient  | $q_2$                     | $U(0.0, 1.0)$      |
| Stellar density             | $\rho_*$ ( $\rho_\odot$ ) | $\rho_* > 0$       |
| GP Parameters               |                           |                    |
| Period                      | $P$ (days)                | $P > 0$            |
| Damping timescale           | $\tau$ (days)             | $\tau > P$         |
| Standard deviation          | $\sigma$                  | $\sigma > 0$       |

**Note.**

<sup>a</sup>  $U(a, b)$  indicates a uniform distribution from  $a$  to  $b$ .

The parameters evolved under uniform priors with only physical limitations (see Table 1). We ran the MCMC using 50 walkers for 100,000 steps and a 20% burn in. The total run was more than 50 times the autocorrelation time, indicating the number of steps was sufficient for convergence.

We present the best-fit parameters in Table 2. We also show the phased-folded light curve and a representative section of the light curve with the GP model and highlighted transit locations in Figure 2.

While we did not fit for eccentricity, we can confirm the orbit is likely near circular from the stellar density fit. The stellar density in the transit fit is affected by transit duration, so discrepancies between the stellar density from the transit fit and the inferred stellar density from the stellar mass and radius would suggest an eccentric orbit (V. Van Eylen & S. Albrecht 2015). The best-fit stellar density from the transit fit ( $1.65^{+0.21}_{-0.54}\rho_\odot$ ) agrees with the inferred stellar density ( $1.696 \pm 0.20\rho_\odot$ ), suggesting our assumption of zero eccentricity did not impact our transit fit.

### 5.1. Transit-timing Variations (TTVs)

Due to TTVs observed in the transits of TOI 2076 b, c, and d (H. P. Osborn et al. 2022; A. I. Lopez Murillo et al. 2025, in preparation), we opted to fit only for the planet parameters of the new candidate; TOI 2076 e. The TTV semiamplitude for b, c, and d are  $\simeq 10$  minutes. Because of the shorter orbital period, we expect the semiamplitude to be smaller in e, allowing us to use a linear ephemeris in our TESS transit fit.

A large TTV signal would cause the transit to smear, artificially increasing the transit duration and decreasing the transit depth. In our fit, this would result in an increased stellar density. The agreement between the best-fit stellar density and inferred density from stellar mass and radius suggests any TTV present in the transits of TOI 2076 e has a minor, or no major, impact on our fit.

We attempted to fit for a TTV amplitude separately from our TESS transit fit. Following the procedure in A. I. Lopez Murillo et al. (2025, in preparation), for each transit, we fit a GP with a SHO kernel to model the local stellar variability simultaneously with a BATMAN transit model in an MCMC framework. We locked all transit parameters except transit midpoint (which was allowed to float within an hour of the

**Table 2**  
Parameters of TOI 2076 e

| Description                                | Parameter                 | Value                                                   |
|--------------------------------------------|---------------------------|---------------------------------------------------------|
| Stellar Parameters                         |                           |                                                         |
| Stellar density                            | $\rho_*$ ( $\rho_\odot$ ) | $1.65^{+0.21}_{-0.54}$                                  |
| Limb-darkening coefficient                 | $q_1$                     | $0.47 \pm 0.13$                                         |
| Limb-darkening coefficient                 | $q_2$                     | $0.218 \pm 0.059$                                       |
| GP Parameters                              |                           |                                                         |
| Standard deviation                         | $\sigma$                  | $0.0214^{+0.0031}_{-0.0024}$                            |
| Period                                     | $P$ (days)                | $3.40^{+0.31}_{-0.26}$                                  |
| Damping timescale                          | $\tau$ (days)             | $3.55^{+0.39}_{-0.31}$                                  |
| Measured Planet Parameters                 |                           |                                                         |
| Time of inferior conjunction               | $T_0$ (BJD-2457000)       | $1740.21306^{+0.00081}_{-0.0008}$                       |
| Orbital period                             | $P$ (days)                | $3.0223445^{+3.3 \times 10^{-6}}_{-3.2 \times 10^{-6}}$ |
| Planet-to-star radius ratio                | $R_P/R_*$                 | $0.0156 \pm 0.001$                                      |
| Impact parameter                           | $b$                       | $0.0 \pm 0.4$                                           |
| Derived Parameters                         |                           |                                                         |
| Semimajor axis to stellar radius ratio     | $a/R_*$                   | $10.4^{+0.43}_{-1.0}$                                   |
| Inclination                                | $i$ ( $^\circ$ )          | $90.0 \pm 2.3$                                          |
| Transit duration (first to fourth contact) | $T_{14}$ (days)           | $0.0902^{+0.0015}_{-0.0013}$                            |
| Planet radius                              | $R_P$ ( $R_J$ )           | $0.1209^{+0.009}_{-0.0087}$                             |
|                                            | $R_P$ ( $R_\oplus$ )      | $1.355^{+0.101}_{-0.098}$                               |
| Semimajor axis                             | $a$ (AU)                  | $0.0385^{+0.0021}_{-0.0049}$                            |
| Equilibrium temperature <sup>a</sup>       | $T_{\text{eq}}$ (K)       | $1138.0^{+72.0}_{-27.0}$                                |

**Notes.**

<sup>a</sup> Assuming zero albedo.

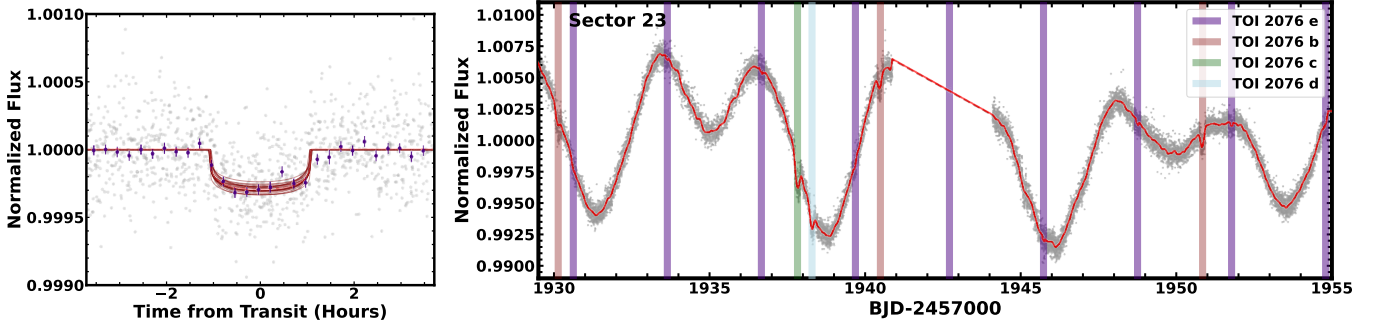
expected midpoint) and allowed the GP parameters to float within physical boundaries, following the same priors as the MISTTBORN GP fit (see Table 1). We found that the individual transits were too shallow to constrain the transit timings well, with fits often not converging on a solution. We were therefore unable to constrain the TTV amplitude.

Future observations at higher SNR will be needed to check for TTV signals in the transits of TOI 2076 e.

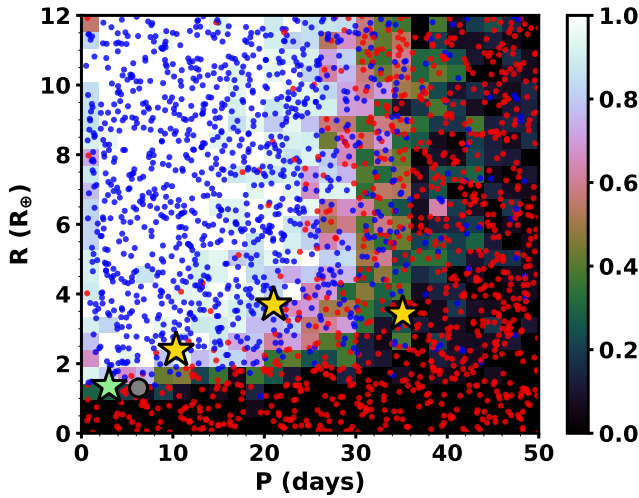
## 6. False-positive Analysis

C. Hedges et al. (2021) used archival Palomar Observatory Sky Survey (R. L. Minkowski & G. O. Abell 1963; I. N. Reid et al. 1991) and PanSTARRS (K. C. Chambers et al. 2016) images, new high-resolution imaging from a range of facilities, and the star's high proper motion ( $\text{pm} = 118.27 \text{ mas yr}^{-1}$ ; Gaia Collaboration et al. 2023) to rule out the presence of a stellar companion as well as a background or foreground star. M. Damasso et al. (2024) gathered more than 300 spectra spanning  $\simeq 3$  yr, which rules out any companion that would be too tight to detect in the imaging (M. L. Wood et al. 2021).

Using the contrast curves from C. Hedges et al. (2021) and the flattened (GP removed) light curve from TESS, we ran TRICERATOPS (S. Giacalone & C. D. Dressing 2020; S. Giacalone et al. 2021) to determine an initial false-positive probability (FPP). TRICERATOPS calculates the probabilities the signal is various transit-like scenarios in a Bayesian framework. We calculate a FPP  $\sim 10^{-5}$ . While this is well



**Figure 2.** (Left) TESS light curve binned to 10 minutes intervals and phase-folded (gray points) with the full (unbinned) TESS light curve phase folded and binned to 15 minutes intervals for clarity (purple points). The best-fit transit model is shown as the bright, opaque red line with 25 model fits pulled from the posterior shown as the dark, translucent red lines. The best-fit GP stellar variability model has been removed from the data and transit model. (Right) Representative section of the TESS light curve (gray points) with the best-fit GP model (red line). Transits of TOI 2076 e are highlighted in purple, and transits of TOI 2076 b, c, and d are highlighted in pink, green, and blue, respectively.



**Figure 3.** Period-radius injection-recovery map for TOI 2076. Blue points indicate recovered signals, while red points mark signals that were not recovered. Only 20% of injected planets are shown for clarity. The background is color coded by the overall completeness in a given bin. The previously known planets in the system are marked as the gold stars, TOI 2076 e is marked as the green star, and candidate TOI 2076 f is marked as the gray circle.

below the threshold for validation, it may be an underestimate of the true FPP because TRICERATOPS requires a flattened light curve and our GP stellar variability model is calculated assuming a transit-like signal is present.

Further evidence supports TOI 2076 e is a real planet:

1. Multi-transiting systems have lower intrinsic (prior) probabilities of being a false positive (J. J. Lissauer et al. 2011; J. F. Rowe et al. 2014; H. Valizadegan et al. 2023).
2. Following A. Vanderburg et al. (2019), we use the transit shape and depth to calculate the faintest companion that could cause the signal to be  $\Delta T < 2.5$  mags. A companion of this magnitude ( $T < 10.8$ ) would have been detected in the suite of imaging or spectra discussed above.
3. While the transit depth is too small to analyze individual transit events, folded transits from each of the three sectors are consistent over a period of four years. Stellar signals would evolve over this period.

Overall, the low false-positive probability based on the transit shape and the suite of follow-up observations confirm that TOI 2076 e is a real planet.

## 7. Injection-recovery Analysis

We performed an injection-recovery analysis to test the detection limits of our pipeline. This is another check of the reliability of our signal (signals in low-completeness zones are more likely to be false alarms) and to determine if there may be other planets residing below our detection limits.

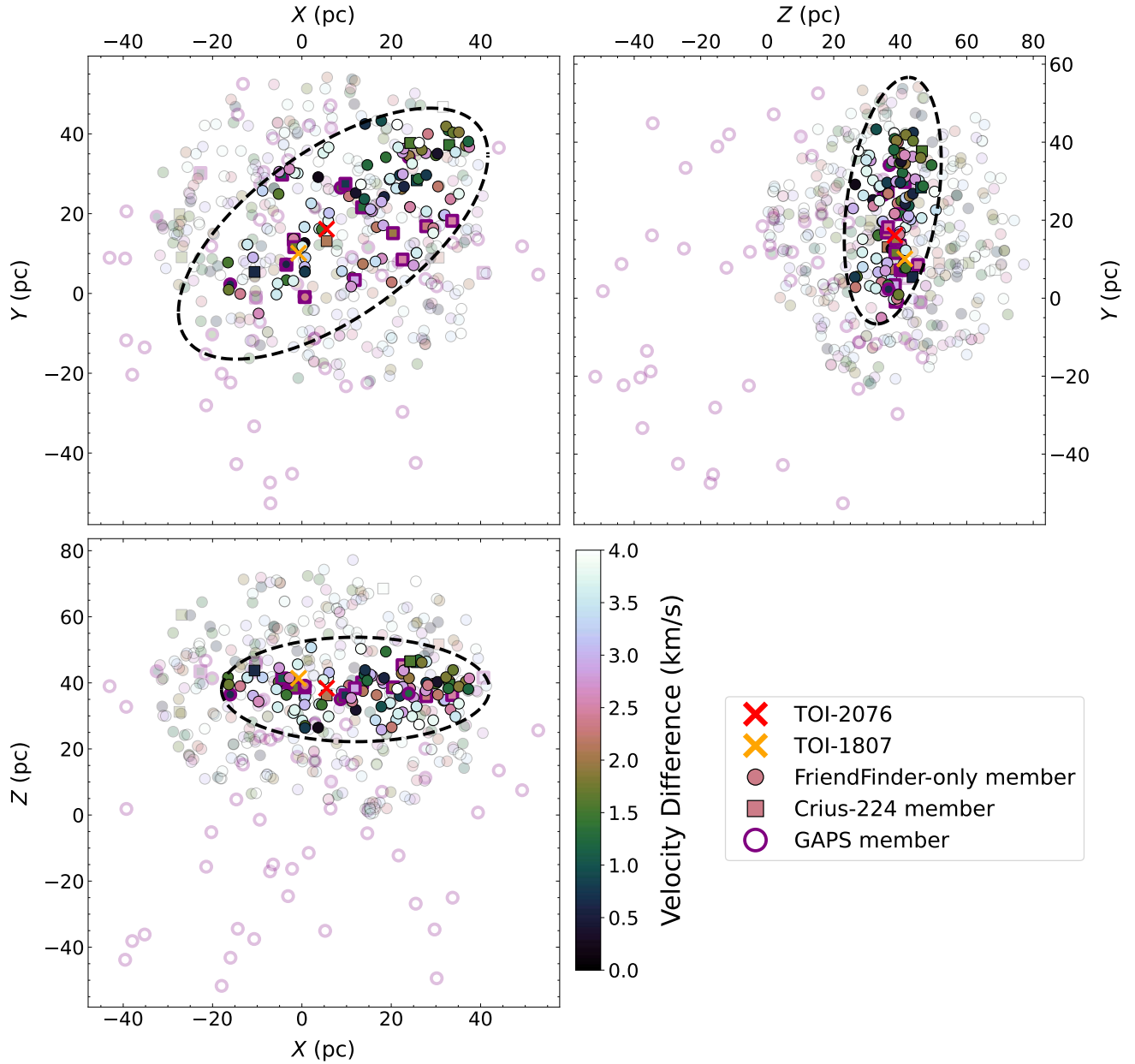
We injected 10,000 synthetic planet signals, following the procedure in A. C. Rizzuto et al. (2017). We randomly generate planets with a period 0.5–50 days, radius 0.01–12  $R_{\oplus}$ , impact parameter 0–1, and  $T_0$  dependent on the period (between the beginning of the data set and the pulled orbital period). We restrict eccentricity to 0 for simplicity. The synthetic planet signal is injected into the raw light curve (as extracted in Section 2), and we attempt to recover the planet by re-running Notch. We show the results in Figure 3. TOI 2076 e falls in a parameter space with an approximately 70% completeness rate, suggesting TOI 2076 e is not likely to be a false alarm.

### 7.1. TOI 2076 f?

The resonance chain of the system as it stands is 3.4:1-2:1-5:3. A planet with a period of approximately 6 days would complete the resonance chain: 2:1-5:3-2:1-5:3. Our pipeline detected an additional signal at 6.25 days with a  $T_0 = 2458740.686$  BJD, which we initially rejected due to low SNR and from visual inspection. If real, this planet would have a radius of  $\sim 1.3$ – $1.4 R_{\oplus}$ . Based on our injection-recovery tests (Section 7) completeness is close to 0% here. Future TESS data or follow-up photometry are needed to confirm or refute this signal.

## 8. Selection of Group Members

TOI 2076 has been previously clustered into groups by the Crius survey (Crius 224, 0.1–0.7 Gyr; L. Moranta et al. 2022) and by the GAPS survey ( $300 \pm 80$  Myr; D. Nardiello et al. 2022). These clusters, however, are small (26 and 76 members, respectively), which can impact the age determination if there are older field interlopers identified as members (A. W. Mann



**Figure 4.** Candidate comoving stars to TOI 2076 (red x) and TOI 1807 (orange x). All stars identified with FriendFinder are shown as colored points. Stars identified as a member of Crius 224 from L. Moranta et al. (2022) are marked as a square, and stars identified as a member in the GAPS survey (D. Nardiello et al. 2022; M. Damasso et al. 2024) are outlined in purple. Stars that fall into all three black, dashed lined regions are used in the age analysis. Stars not used in the analysis but identified as candidate members are shown in lower opacity.

et al. 2022). For simplicity, we will refer to the population as Crius 224 for the rest of the paper.

We searched for additional comoving stars in Crius 224 using FriendFinder<sup>7</sup> (B. M. Tofflemire et al. 2021). To summarize, FriendFinder uses Gaia Data Release 3 astrometry and an input radial velocity of the star of interest to compute XYZ positions and UVW velocities of nearby stars. FriendFinder then looks for stars with tangential velocities and XYZ coordinates within user-set bounds of the target.

We initially restricted our selection to stars with a tangential velocity ( $V_{\text{tan}}$ ) within  $4 \text{ km s}^{-1}$  and a 3D distance within 40 pc.

This resulted in a list of 450 comoving stars, 402 of which were unique to the FriendFinder list. We combined all three sets of stars, FriendFinder, Crius 224, and the GAPS survey, to create a list of 489 candidate members.

There was a clear string of over density of nearby, comoving stars in XYZ space but a large portion of contaminating stars in the field. Our goal is to derive a precise age for Crius 224/TOI 2076, for which we want a clean (rather than complete) list, so we applied elliptical cuts designed to encompass the over density of stars (shown in Figure 4). The parameters of these ellipses are somewhat arbitrary, and many true members will be more distant. However, inspection of the light curves of stars outside these cuts suggests there are far more interlopers (slow rotators) than true members (stars showing clear young-star

<sup>7</sup> <https://github.com/adamkraus/Comove>

variability); the opposite is true inside the ellipses. The ellipse cuts also provided a sufficient list of likely members for a precise age determination.

Our final membership list includes 125 stars. We list the candidate members in Table 3. In addition to TOI 2076, the membership list includes TOI 1807, which was first seen to be comoving with TOI 2076 in C. Hedges et al. (2021).

## 9. Update Group Age

In order to derive a precise age for TOI 2076’s parent population, we combine the age determinations using four methods: gyrochronology, Li depletion, isochronal modeling, and variability-based aging.

### 9.1. Gyrochronology

As stars age, their rotation slows due to the loss of angular momentum from magnetized stellar winds. This phenomenon implies that rotation rates can serve as indicators of stellar age—a method known as gyrochronology (S. A. Barnes 2007; J. L. van Saders et al. 2016).

Using the unpoplar package (S. Hattori et al. 2022), we generated TESS light curves for the 125 stars in the final membership list. We measured rotation periods using a Lomb–Scargle periodogram (N. R. Lomb 1976; J. D. Scargle 1982) with a linearly spaced search grid of 100,000 steps spanning 0.2–20 days. The rotation period corresponding to the highest Lomb–Scargle power is adopted as the measured period. For stars observed in multiple sectors, we ran the Lomb–Scargle on each sector separately and adopted the period from the sector with the highest measured power. Uncertainties in the rotation measurements are estimated using the empirical uncertainty relation from A. W. Boyle et al. (2025).

Given TESS’s large 21'' pixel size, close binary companions may introduce spurious rotation signals into the target star’s light curve, potentially leading to incorrect period measurements. To minimize this effect, stars with a Gaia RUWE > 1.2 or flagged as `non_single_star == 1` were excluded.

Each star’s light curve was manually inspected to validate the period measurement. Light curves are categorized as “good” (a clear rotation signal with a period visually consistent with the Lomb–Scargle result), “fair” (a less distinct rotation signal or noisy light curve), “poor” (no visible rotation or an evidently incorrect period), or “binary” (multiple superimposed signals). To ensure reliable gyrochronological age derivations, only measurements classified as “good” are included (Figure 5).

We find the rotation period of TOI 2076 to be  $P_{\text{rot}} = 7.36 \pm 0.30$  days with a “good” quality. This is consistent with the value found in D. Nardiello et al. (2022;  $7.29 \pm 0.12$  days) and C. Hedges et al. (2021;  $7.27 \pm 0.23$  days).

The rotation-based age of the group is calculated using `gyro-interp` (L. G. Bouma et al. 2023), which requires the star’s rotation period (and its uncertainty) and effective temperature (and its uncertainty) as inputs. Effective temperatures are derived by dereddening each star’s Gaia DR2  $G_{\text{BP}} - G_{\text{RP}}$  colors using the STILISM dust maps (R. Lallement et al. 2019), followed by conversion to effective temperatures using the empirical color–temperature relation from J. L. Curtis et al. (2020).

We generated the age posterior distribution for each star with `gyro-interp`. To infer the group’s age, the posteriors

are combined using `PosteriorStacker` (L. Baronchelli et al. 2020), which assumes a Gaussian intrinsic age distribution. This procedure yields a final gyrochronology age for the group of  $238^{+69}_{-60}$  Myr.

### 9.2. Lithium Sequence

The lithium (Li) sequence is a method of determining the age of an association based on the observed Li levels of stellar members (e.g., J. D. Cummings et al. 2017; R. D. Jeffries et al. 2017; M. L. Wood et al. 2023b). Li is destroyed in stars with core temperatures greater than  $2.5 \times 10^6$  K. Different stellar masses reach this interior temperature at different rates, and mixing can deplete Li levels in the photosphere, leading to a relationship between age and observed Li levels relying on the boundary between stars which have burned their Li and stars that have not (e.g., A. S. Binks & R. D. Jeffries 2014; A. S. Binks et al. 2021).

We drew Li equivalent-width (EW) measurements from D. Nardiello et al. (2022) for eight stars (Figure 6). Using `EAGLES` (R. D. Jeffries et al. 2023), we estimated the cluster’s age based on the Li EW of the candidate comoving members. `EAGLES` requires the stars’ effective temperatures (rather than color), so we converted the Gaia DR2  $G_{\text{BP}} - G_{\text{RP}}$  measurements to  $T_{\text{eff}}$  following the same methodology as in Section 9.1.

This resulted in an age of  $210^{+45}_{-37}$  Myr.

### 9.3. Isochronal Modeling

We compared the Gaia color–magnitude diagram of our updated membership list to PARSECv1.2 isochrones (A. Breßan et al. 2012). For this comparison, we used a Gaussian mixture model following the appendix of A. W. Mann et al. (2022) and M. L. Wood et al. (2023a). To briefly summarize, we used a likelihood formed from the mixture of two components. The first described a single-age single-star sequence described by age ( $\tau$ ) and extinction ( $E(B - V)$ ); metallicity is assumed to be solar. The second component describes the outliers (e.g., nonmembers or binaries). Since our membership list is already relatively clean of nonmembers, we forced the second component to behave like a binary sequence, modeled as an offset from the single-star sequence and a variance ( $Y_B$  and  $V_B$ ). Two additional free parameters describe the amplitude of the outlier model ( $P_B$ ) and underestimated uncertainties in the model and/or photometry ( $f$ ). The variables  $Y_B$ ,  $V_B$ ,  $f$  are all measured in magnitudes.

We performed the fit using a MCMC framework with `emcee` (D. Foreman-Mackey et al. 2013). We ran the fit using 50 walkers for 50,000 steps after an initial burn in of 10,000 steps. Because all stars are within the Local Bubble, we placed a weak Gaussian prior on extinction ( $0 \pm 0.1$  mag); all other parameters evolved under uniform priors.

The final fit gave an age of  $\tau = 195 \pm 21$  Myr (Figure 7). Letting metallicity float within near-Solar values ( $-0.3 < [M/H] < +0.3$ ) yielded a less precise but similar  $\tau = 197 \pm 26$  Myr and still favored a near-Solar metallicity ( $0.05 \pm 0.10$ ). We adopt the latter as our isochronal age.

Most groups at this age and mass do not have a clear population of pre-main-sequence stars detected by Gaia, and they are not massive enough to have a large population of turn-off stars. Prior such analyses have yielded a nearly uniform age from 150–800 Myr (P. C. Thao et al. 2024). In this case, the



**Table 3**  
Candidate Co-moving Stars With TOI 2076

| Gaia DR3            | $\alpha$<br>(J2016) | $\delta$<br>(J2016) | Gmag<br>(mag) | $B_p-R_p$<br>(mag) | $V_{off}$<br>(km s <sup>-1</sup> ) | $RV_{exp}^a$<br>(km s <sup>-1</sup> ) | $RV^b$<br>(km s <sup>-1</sup> ) | $\sigma_{RV}^b$<br>(km s <sup>-1</sup> ) | $P_{rot}$<br>(days) | $\sigma_{Prot}$<br>(days) | $Q^c$ | $Li^d$<br>(mÅ) | $\sigma_{Li}^d$<br>(mÅ) | FF <sup>e</sup> | Crius <sup>f</sup> | GAPS <sup>g</sup> | Final <sup>h</sup> |
|---------------------|---------------------|---------------------|---------------|--------------------|------------------------------------|---------------------------------------|---------------------------------|------------------------------------------|---------------------|---------------------------|-------|----------------|-------------------------|-----------------|--------------------|-------------------|--------------------|
| 1490845584382687232 | 217.392             | 39.7904             | 8.92          | 1.03               | 0.0                                | 0.0                                   | -13.08                          | 0.15                                     | 7.36                | 0.3                       | g     | 89.4           | 3.0                     | Y               | Y                  | Y                 | Y                  |
| 1500434116051163904 | 207.3735            | 41.627              | 16.60         | 4.48               | 0.18                               | 0.18                                  | ...                             | ...                                      | 1.98                | 0.02                      | p     | ...            | ...                     | Y               | N                  | N                 | Y                  |
| 1426873554041125120 | 247.76              | 54.227              | 14.40         | 2.96               | 0.2                                | 0.2                                   | -26.13                          | 4.47                                     | 0.74                | 0.0                       | p     | ...            | ...                     | Y               | N                  | N                 | Y                  |
| 1311701639472895616 | 248.9159            | 29.0931             | 16.04         | 3.44               | 0.36                               | 0.36                                  | ...                             | ...                                      | 2.57                | 0.04                      | p     | ...            | ...                     | Y               | N                  | N                 | Y                  |
| 1383172330523940096 | 240.5723            | 40.4968             | 12.55         | 2.36               | 0.37                               | 0.37                                  | -18.28                          | 1.62                                     | 2.44                | 0.03                      | g     | ...            | ...                     | Y               | N                  | N                 | Y                  |
| 1476485996883837184 | 201.2826            | 38.9224             | 9.68          | 1.26               | 0.38                               | 0.38                                  | -7.14                           | 0.15                                     | 9.01                | 0.45                      | f     | 104.0          | 1.0                     | Y               | Y                  | Y                 | Y                  |
| 1370571450529693952 | 234.4453            | 33.1002             | 14.27         | 2.77               | 0.56                               | 0.56                                  | -17.89                          | 2.45                                     | 1.07                | 0.01                      | g     | ...            | ...                     | Y               | N                  | N                 | Y                  |
| 1523644978312478976 | 194.6451            | 38.2799             | 9.03          | 1.18               | 0.58                               | 0.58                                  | -3.34                           | 6.85                                     | 5.17                | 0.15                      | g     | 111.9          | 1.1                     | Y               | N                  | Y                 | Y                  |
| 1321207948287911168 | 237.3212            | 30.9853             | 14.57         | 2.81               | 0.6                                | 0.6                                   | ...                             | ...                                      | 0.76                | 0.0                       | g     | ...            | ...                     | Y               | N                  | N                 | Y                  |
| 1535849763538623232 | 182.9662            | 39.9578             | 11.01         | 1.7                | 0.62                               | 0.62                                  | -2.39                           | 0.58                                     | 1.75                | 0.02                      | g     | ...            | ...                     | Y               | Y                  | N                 | Y                  |

**Notes.** Table 3 is published in its entirety in the electronic edition of the Astrophysical Journal. A portion is shown here for guidance regarding its form and content.

<sup>a</sup> Expected radial velocity for membership.

<sup>b</sup> From GaiaDR3 (Gaia Collaboration et al. 2023).

<sup>c</sup> Indicates reliability of the rotation period as described in Section 9.1; g = good, f = fair, p = poor, b = binary.

<sup>d</sup> From D. Nardiello et al. (2022).

<sup>e</sup> Indicates source was identified as comoving with TOI 2076 with FriendFinder.

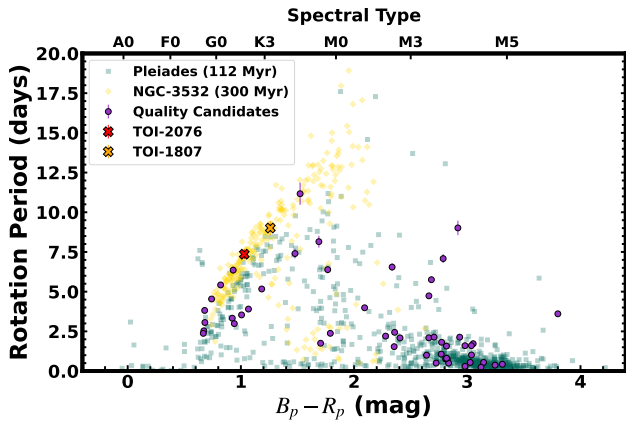
<sup>f</sup> Indicates source was identified as comoving with TOI 2076 in Crius 224.

<sup>g</sup> Indicates source was identified as comoving with TOI 2076 in the GAPS survey.

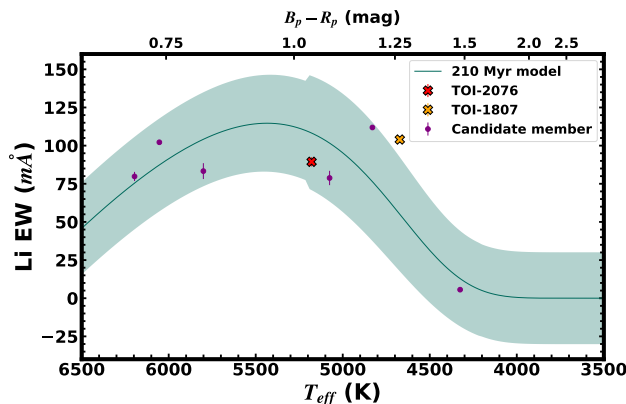
<sup>h</sup> Indicates source was used in the age analysis.

(This table is available in its entirety in machine readable form in the [online article](#).)





**Figure 5.** The rotation periods of the candidate comoving stars (purple circles) against color (as a proxy for stellar type). Only the high-quality, high-reliability rotation periods are shown. TOI 2076 is marked as the red x, and TOI 1807 is marked as the orange x. The rotation sequence for Pleiades (green squares; L. M. Rebull et al. 2016) and NGC 3532 (yellow diamonds; D. J. Fritzewski et al. 2021) are shown for reference. The stars comoving with TOI 2076 show a distribution in between Pleiades and NGC 3532, consistent with the derived gyrochronology age.



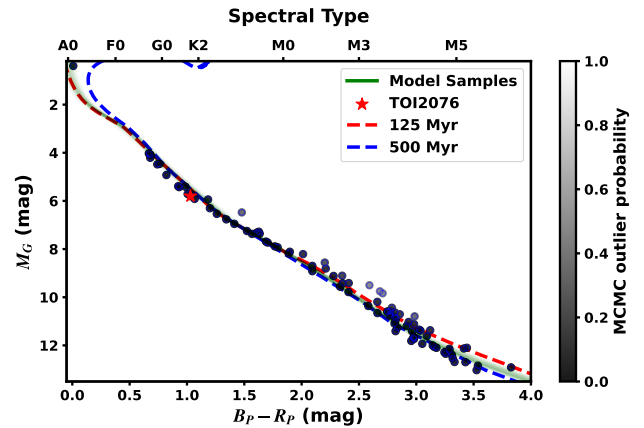
**Figure 6.** The Li equivalent widths of the candidate comoving stars against effective temperature (as a proxy for color; purple points). TOI 2076 is shown as the red x, and TOI 1807 is shown as the orange x. The teal region shows the expected distribution for the best-fit age (210 Myr).

constraints on the upper age end rely mostly on a single A0V star (HD 153808 or  $\epsilon$  Her).

HD 153808 (Gaia DR3 1310284059747409920) has a radial velocity of  $-23$  to  $-25$  km s $^{-1}$  (A. Tokovinin 2018; Gaia Collaboration et al. 2023), strongly indicating it is a real member (perfect agreement is  $-23.4$  km s $^{-1}$ ). The star is a spectroscopic binary (D. Pourbaix et al. 2004), but the CMD is predicted to be nearly vertical in this range (Figure 7), so this does not impact the analysis. Prior analyses of HD 153808 have also yielded similar ages (150–230 Myr; G. H. Rieke et al. 2005; T. J. David & L. A. Hillenbrand 2015). This also agrees with all our other age constraints. Thus, we opt to include the isochronal age in our group age analysis.

#### 9.4. Variability

M. G. Barber & A. W. Mann (2023) developed a method of estimating ages of young stellar clusters using the excess uncertainties in Gaia photometry (M. Riello et al. 2021). The method takes advantage of more variable sources having higher photometric uncertainties than their quiet counterparts to fit a Skumanich-like relation between stellar activity and



**Figure 7.** Color–magnitude diagram of likely members of Cris 224 (black circles) including TOI 2076 (red star) compared to isochrones from PARSEC. The green lines are random samples from the MCMC posterior. Circles are colored by their outlier probability (mixture weight). We show 125 Myr (red dashed) and 500 Myr (blue dashed) lines for reference. The lack of pre-main-sequence M dwarfs rules out ages  $\lesssim 150$  Myr, while higher ages are ruled out by the A star HD 153808.

age. EVA<sup>8</sup> (Excess Variability-based Age) queries Gaia, makes appropriate stellar cuts, and fits for the age. Inputting the list derived in Section 9, EVA calculated an age of  $252^{+121}_{-82}$  Myr taking into account all three bands;  $G$ ,  $G_{BP}$ , and  $G_{RP}$ . The variability-based age is robust against field interlopers but the uncertainties are strongly affected by the number of stars in the sample. Given our group is only 125 stars, the large uncertainties are unsurprising.

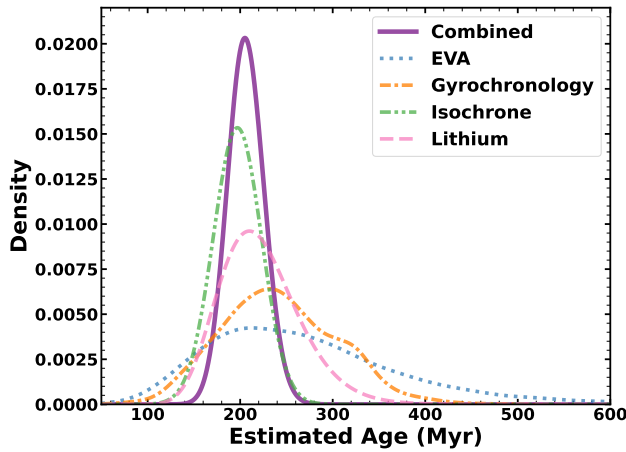
#### 9.5. Combining Age Determinations

We combine the individual age determinations for Cris 224 by taking into account the uncertainties calculated in each estimate. The estimates are not truly independent; the gyrochronological, lithium, and variability ages are calibrated using similar clusters and there are known relations between gyrochronological and lithium ages (e.g., J. Bouvier et al. 2018). However, each estimate is limited by random uncertainties and the methods contain a mix of model-based methods (isochrone fitting) and empirical ones (gyrochronological and lithium), and rely on different sets of stars (e.g., an isochrone is mostly from the A stars while rotation and lithium mostly comes from FGK stars). Further, even if the absolute age uncertainty is underestimated, the combined age will be robust in a relative sense (compared to the calibration clusters).

Taking into account each estimate’s likelihood distribution (see Figure 8), we find an overall age of  $210 \pm 20$  Myr for Cris 224. This estimate more closely agrees with the results of C. Hedges et al. (2021), and is consistent with M. Damasso et al. (2024) to  $\simeq 1\sigma$  and H. P. Osborn et al. (2022) to  $2\sigma$ . Our value is also far more precise than prior determinations, with uncertainties in line with similar-aged associations (S. Meibom et al. 2011; P. C. Thao et al. 2024).

Numerous studies have found evidence of age spreads in young clusters, often in the form of an extended main-sequence turn-off (e.g., S. Gossage et al. 2019) or a spread in the luminosity of pre-main-sequence stars (e.g., L. Cao et al. 2022). Our quoted uncertainty would depend on both measurement uncertainties and any age spread, as such effects

<sup>8</sup> <https://github.com/madysonb/EVA>



**Figure 8.** The final age estimate distribution ( $210 \pm 20$  Myr; solid purple) compared to the age estimates from each method (colored lines).

would broaden the posteriors for any of the methods above. However, these spreads are usually small ( $<5$  Myr; R. D. Jeffries et al. 2011; S. Ratzenböck et al. 2023) compared to the 20 Myr uncertainty here, and there is no evidence of subgroups in the kinematics of Crius 224. Thus, we attribute most or all of the 20 Myr uncertainty to measurement and model limitations for measuring the age. Age spreads might be detectable in such groups with significantly more data.

## 10. Known Planets in Crius 224

We searched for additional planet systems in Crius 224 by crossmatching our initial (preellipsoid cut) candidate comoving stars list against the known planets and TOI list. We identify planet-hosting stars overlapping in the field, some of which may be true members. We discuss individual systems below, organized by detection method.

### 10.1. Transiting Systems

TOI 1807 (TIC 180695581, Gaia DR3 1476485996883837184; C. Hedges et al. 2021) was previously determined to be a member, which we confirm here. The rotation period and lithium are consistent with a similar age, and the six-dimensional coordinates (XYZUVW) are within the core of the group (Figure 4).

TOI 1266 marginally falls outside of our ellipsoidal cuts and exhibits a slow rotation period ( $>20$  days). There are no other signs of youth (G. Stefánsson et al. 2020), so we conclude this star is not a member.

### 10.2. Directly Imaged Systems

G 196-3 (Gaia DR3 824017070904063104; K.-U. Michel & M. Mugrauer 2024) is a directly imaged system that lands slightly outside the ellipsoidal bounds in Z, and has a modest tangential velocity offset ( $3.5 \text{ km s}^{-1}$ ) from the core of the group. We find a short rotation period ( $1.31 \pm 0.01$  days) consistent with Crius 224. However, the host is an M3V; such cool stars can maintain fast rotation out to  $\approx 600$  Myr (L. G. Bouma et al. 2023). We consider this a potential, but unlikely member. If G 196-3 is a member, this increases the age. Prior determinations place G 196-3 at 55–135 Myr by UV emission (R. Galicher et al. 2016) and 10–150 Myr based on the Li EW of the stellar companion (G 196-3 B; J. D. Kirkpatrick et al. 2008; K. N. Allers & M. C. Liu 2013).

### 10.3. Radial Velocity Systems

HD 168746 (Gaia DR3 4153637759337630720; F. Pepe et al. 2002) did not make our list, but was in the original Crius 224 list from L. Moranta et al. (2022). HD 98736 (Gaia DR3 3977050728669174912; K. Ment et al. 2018) and HIP 55507 (Gaia DR3 788340461203094912; F. Feng et al. 2022) both make the FriendFinder list. All three systems sit outside the ellipsoidal boundaries and have only marginal evidence of membership. The latter two both have RVs  $6\text{--}10 \text{ km s}^{-1}$  from the core of Crius 224; too far to be explained by random motion or measurement errors. HD 168746 has a rotation period consistent with membership ( $9.30 \pm 0.49$  days,  $B_p - R_p = 0.86$ ), but this period is considered unreliable due to having only one sector of data.

## 11. Additional Candidate Planets as Candidate Group Members

We searched the full list of identified comoving members for transiting planets. Of the 489 stars in our initial group list, we exacted light curves and searched for planets for 435 stars following the methodology described in Sections 2.1 and 3, adjusting the notch filtering window based on the rotation period of the star.

No signal was both a clear group member and a clear planetary signal. We discuss individual signals and host stars below. We summarize the results in Table 4.

### 11.1. Planet Candidates

We recover a transit signal in TIC 8673878 (Gaia DR3 1324251396473528832) with a period of 16.01 days,  $T_0 = 1971.46$  TESS-BJD, and depth  $\sim 0.5\%$ . Due to data gaps, this results in only three detected transits, the last one with a flare shortly following egress (poor quality). TIC 8673878 sits inside of our ellipsoidal cuts. It has a rotation period consistent with membership for its color ( $P_{\text{rot}} = 7.08$  days,  $B_p - R_p = 2.79$ ) and a tangential velocity within  $3 \text{ km s}^{-1}$  of expected. TIC 8673878 is a likely true member of the group, but the quality of the transits is insufficient to confirm this as a planet.

TIC 17563961 (Gaia DR3 1329910067425244160) is a known wide binary system (2065.1 au; H.-J. Tian et al. 2020). Around the primary, we recover a signal with a period of 16.29 days,  $T_0 = 1972.01$  TESS-BJD, and depth  $\sim 0.05\%$ . We recover three consecutive transits in the first two sectors of TESS data available, but transits in the next two sectors all fall in data gaps. Only one signal is recovered in the last sector, and this falls on the edge of a data gap. Both FriendFinder and L. Moranta et al. (2022) put TIC 17563961 into Crius 224. The star exhibits a fast rotation period ( $P_{\text{rot}} = 2.37$  days) and tangential velocity within  $2 \text{ km s}^{-1}$  of expected. TIC 17563961 is a likely true member of the group, but the transit is too shallow and has too few transits to confirm.

In TIC 68033619 (Gaia DR3 1488431297366254464), we recover a  $\sim 0.3\%$  signal with  $T_0 = 1744.07$  TESS-BJD and a period of 26.55 days. However, we do not recover consecutive transits so this period may be an alias or a false positive. TIC 68033619 was grouped with TOI 2076 with FriendFinder and sits inside the ellipsoidal bounds. The rotation period is short ( $P_{\text{rot}} = 2.09$  days), but the radial velocity is  $>10 \text{ km s}^{-1}$  off from expected. TIC 68033619 is unlikely to be a true member of the group.

**Table 4**  
Candidate Signals Identified in our Initial, Uncut List of 489 Comoving Stars With TOI 2076

| TIC                     | Gaia DR3            | $T_0$<br>(BJD-2457000) | Period<br>(days)    | Approx. Depth<br>(%) | Member? |
|-------------------------|---------------------|------------------------|---------------------|----------------------|---------|
| Planet Candidates       |                     |                        |                     |                      |         |
| 8673878                 | 1324251396473528832 | 1971.46                | 16.01               | 0.5                  | Y       |
| 17563961                | 1329910067425244160 | 1972.01                | 16.29               | 0.05                 | Y       |
| 68033619                | 1488431297366254464 | 1744.07                | 26.55 <sup>a</sup>  | 0.3                  | N       |
| 400231203               | 4549900100674763264 | 1984.52                | 27.78               | 0.02                 | N       |
|                         |                     | 1986.40                | 16.55 <sup>a</sup>  | 0.02                 | N       |
| Other Object Candidates |                     |                        |                     |                      |         |
| 16271601                | 1278340773059748608 | 2714.21                | 721.38 <sup>a</sup> | 12                   | Y       |
| 10150705                | 767786671869455360  | 1899.57                | 1.09                | 0.07                 | N       |
| 392909786               | 3714227557974504576 | 1928.22                | 0.204               | 60                   | ?       |

**Note.**

<sup>a</sup> Likely alias.

We recover a shallow (depth  $\sim 0.02\%$ ), 27.78 day signal with  $T_0 = 1984.52$  TESS-BJD in TIC 400231203 (Gaia DR3 4549900100674763264). Only three transits are visible across the five sectors. We also recover a signal with a similar depth ( $\sim 0.02\%$ ),  $T_0 = 1986.40$  TESS-BJD, and a period of 16.55 days, though visual inspections suggest this is likely an alias of the true period. TIC 400231203 was identified as a candidate group member with FriendFinder, though it falls outside of our ellipsoidal cuts. TIC 400231203 exhibits a fast rotation period ( $P_{\text{rot}} = 4.03$  days), though with a poor quality and visual inspection suggests a much longer rotation period. The radial velocity is  $> 10 \text{ km s}^{-1}$  off from expected for membership. TIC 400231203 is not a likely member of the group.

### 11.2. Other Object Candidates

We identify two transit-like events in the light curve of TIC 16271601 (Gaia DR3 1278340773059748608) at 2714.21 and 3435.59 TESS-BJD. Due to the sector breaks, we cannot recover a period for the system. The single transit at 2714.21 was previously reported as a Community TOI.<sup>9</sup> The signals are deep ( $\sim 12\%$ ) and V-shaped, making it a likely eclipsing binary system. TIC 16271601 falls within the ellipsoidal cuts. The rotation period ( $P_{\text{rot}} = 0.5$  day) is consistent with group membership, making TIC 16271601 a likely true group member.

TIC 10150705 (Gaia DR3 767786671869455360) is a previously reported spectroscopic binary (D. Pourbaix et al. 2004). We recover a transit-like signal with a period of 1.09 day and  $T_0 = 1899.57$  TESS-BJD, consistent with the TCE reported for the system for Sector 48.<sup>10</sup> The signal exhibits odd-even differences, consistent with its binary status. TIC 10150705 falls outside the ellipsoidal bounds. The light curve exhibits a rotation period of  $P_{\text{rot}} = 5.64$  days, slightly slow for membership for its color ( $B_p - R_p = 0.76$ ), but has a radial velocity  $> 10 \text{ km s}^{-1}$  from expected, though this may be due to its binary status. TIC 10150705 is unlikely a true member of the group.

TIC 392909786 (Gaia DR3 3714227557974504576) has TESS TCE reports with a period of 0.408 day and a period of

0.204 day. It is a known eclipsing binary (T. Pribulla et al. 2003; A. Prša et al. 2022), with a true period of 0.204 day. Due to the short period, we do not recover the signal in our pipeline (our minimum search period is 0.5 day). TIC 392909786 falls outside our ellipsoidal bounds. Our rotation period pipeline identified the binary signal as the rotation period, and we were unable to estimate the true rotation period through visual inspection. The close binary orbit also makes radial velocities challenging, with Gaia unable to recover a radial velocity for the star, hindering our ability to confirm or dispute membership.

## 12. Summary and Discussion

We report the discovery of TOI 2076 e, a super-Earth interior to the three previously discovered sub-Neptune-sized planets in the system. We find TOI 2076 e to be  $1.35 R_{\oplus}$  on a 3.022 days orbit. We rederive the age of the system using comoving stars in the field and previous lists of identified potential cluster members. Using rotations, lithium levels, isochronal modeling, and variabilities, we find the parent population of TOI 2076 (Crius 224) to be  $210 \pm 20$  Myr and note other planets and planet candidates that likely reside within the association.

TOI 2076 is a target of the Keys to Revealing the Origin and Nature Of sub-Neptune Systems JWST program (GO 5959; A. Feinstein et al. 2024) focusing on the atmospheric compositions of young sub-Neptunes in multiplanet systems. TOI 2076 is also of interest because of significant TTVs, which may eventually yield masses and eccentricities for the planets (H. P. Osborn et al. 2022). The previously unknown planet could effect the interpretation of the system, both the TTVs (e.g., D. Weissman et al. 2023) and as part of a comparison between planets in this system. The updated age may also be important for the interpretation of the masses and atmospheres of all four planets.

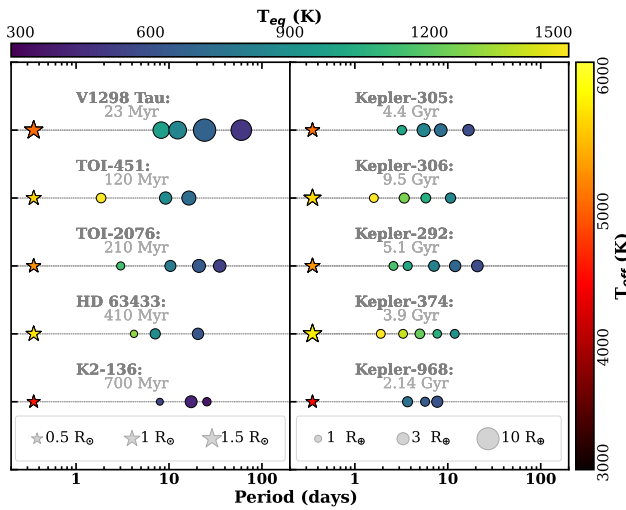
TOI 1807 is part of two JWST programs; one searching for silicate vapor atmospheres (Cycle 3, GO 4818; M. Weiner Mansfield et al. 2024), and one studying the evolution of lava worlds (Cycle 4, GO 8864; L. Dang et al. 2025). Similar to TOI 2076, the updated age may be important for interpreting the atmosphere and the evolution of such lava worlds.

As we show in Figure 1, our custom light curve extraction is capable of detecting smaller planets than the default extraction using the PDCSAP pipeline. This is a more extreme example

<sup>9</sup> <https://exofop.ipac.caltech.edu/tess/target.php?id=tic16271601>

<sup>10</sup> [https://exo.mast.stsci.edu/exomast\\_planet.html?planet=TIC10150705S0048S0048TCE1](https://exo.mast.stsci.edu/exomast_planet.html?planet=TIC10150705S0048S0048TCE1)





**Figure 9.** Young, multiplanet transiting systems found in a stellar cluster or association (left) compared to mature multiplanet systems from Kepler around similar host stars (right). The mature  $\geq 3$ -planet systems show a higher level of intrasystem uniformity in period and radius space than the young systems. The inclusion of TOI 2076 e pushes the system out of uniformity.

than typical, and the SAP flux for the sector does not contain the  $\sim 0.5$  day spurious signal. However, the PDCSAP flux is commonly used in young planet searches (e.g., E. R. Newton et al. 2021; A. W. Mann et al. 2022; S. Vach et al. 2022; M. L. Wood et al. 2023a), highlighting the importance of light curve extraction in planet search sensitivity. In M. G. Barber et al. (2024b), we were similarly unable to detect the additional transiting planet in the PDCSAP flux to a high enough SNR in our search pipeline despite a strong signal using our custom extraction. These discoveries highlight the need of not only sensitive search pipelines but careful handling of data extraction as well.

The mature Kepler multiplanet transiting population displays a higher level of intrasystem uniformity in period and radius space than the young TESS multiplanet systems (J. J. Lissauer et al. 2011; R. I. Dawson et al. 2016; L. M. Weiss et al. 2018). With only a limited sample of young,  $\geq 2$ -planet systems, it is unclear if this is true for young planets, especially given detection bias and decreased sensitivity to young planets largely from K2 and TESS systems in comparison to mature planets seen by Kepler.

The discovery of TOI 2076 e hints that detection bias is making young systems appear more uniform. Among the young  $\geq 3$ -planet systems, TOI 2076 was the only system to show a high level of intrasystem uniformity comparable to the mature Kepler planets (Figure 9). The detection of the smaller, inner planet pushes the system out of radius and period uniformity.

F. Dai et al. (2024) and J. H. Hamer & K. C. Schlaufman (2024) note an excess of young planetary systems in mean motion resonance (MMR). While TOI 2076 e is not in MMR with neighboring TOI 2076 b (3.4:1), TOI 2076 e is just outside a 7:1 MMR with TOI 2076 d (6.95:1). TOI 2076 b and c (2.03:1) and c and d (1.67:1, or 5.01:3) are near MMR as well. A fifth planet with an orbital period near 6 days would complete the resonance chain 2:1-5:3-2:1-5:3. While our candidate signal at 6.25 days does not pass our quality thresholds, future TESS data could be used to search for missing planet f.

The discovery of TOI 2076 e is motivation to revisit previously identified transiting systems to look for additional

signals. With a more complete search and a good understanding of our completeness (e.g., from injection/recovery) we can control for biases intrinsic to young stars and map out evolutionary effects like intrasystem uniformity. As our survey continues, we expect to find longer period planets due to additional data, but we may also become more sensitive to smaller planets like TOI 2076 e.

## Acknowledgments

The authors thank Halee, Bandit, and Elsa for their invaluable support during the creation of the manuscript. M.G.B. was supported by the NSF Graduate Research Fellowship (DGE-2040435) and the TESS Guest Investigator Cycle 6 program (22-TESS22-0013). A.W.M. was supported by the NSF CAREER program (AST-2143763) and a grant from NASA’s Exoplanet Research Program (XRP 80NSSC21K0393). A.W.B. was supported by the NSF Graduate Research Fellowship (DGE-2439854) and funding from NASA’s Astrophysics Data Analysis program (ADAP; 80NSSC24K0619). A.I.L.M. was supported by the Chancellor’s Science Scholars Program.<sup>11</sup>

Funding for the TESS mission is provided by NASA’s Science Mission Directorate. We acknowledge the use of public TESS data from pipelines at the TESS Science Office and at the TESS Science Processing Operations Center. Resources supporting this work were provided by the NASA High-End Computing (HEC) Program through the NASA Advanced Supercomputing (NAS) Division at Ames Research Center for the production of the SPOC data products. TESS data presented in this paper were obtained from the Mikulski Archive for Space Telescopes (MAST) at the Space Telescope Science Institute.

*Facility:* TESS.

*Software:* MISTTBORN, FriendFinder, Notch & LOCoR.

## ORCID iDs

Madysen G. Barber <https://orcid.org/0000-0002-8399-472X>  
 Andrew W. Mann <https://orcid.org/0000-0003-3654-1602>  
 Andrew Vanderburg <https://orcid.org/0000-0001-7246-5438>  
 Andrew W. Boyle <https://orcid.org/0000-0001-6037-2971>  
 Ana Isabel Lopez Murillo <https://orcid.org/0009-0006-9572-1733>

## References

- Allers, K. N., & Liu, M. C. 2013, *ApJ*, **772**, 79
- Barber, M. G., & Mann, A. W. 2023, *ApJ*, **953**, 127
- Barber, M. G., Mann, A. W., Vanderburg, A., et al. 2024a, *Natur*, **635**, 574
- Barber, M. G., Thao, P. C., Mann, A. W., et al. 2024b, *ApJL*, **973**, L30
- Barnes, S. A. 2007, *ApJ*, **669**, 1167
- Baronchelli, L., Nandra, K., & Buchner, J. 2020, *MNRAS*, **498**, 5284
- Binks, A. S., & Jeffries, R. D. 2014, *MNRAS*, **438**, L11
- Binks, A. S., Jeffries, R. D., Jackson, R. J., et al. 2021, *MNRAS*, **505**, 1280
- Blunt, S., Carvalho, A., David, T. J., et al. 2023, *AJ*, **166**, 62
- Bouma, L. G., Palumbo, E. K., & Hillenbrand, L. A. 2023, *ApJL*, **947**, L3
- Bouvier, J., Barrado, D., Moraux, E., et al. 2018, *A&A*, **613**, A63
- Boyle, A. W., Mann, A. W., & Bush, J. 2025, *ApJ*, **985**, 233
- Bressan, A., Marigo, P., Girardi, L., et al. 2012, *MNRAS*, **427**, 127
- Brewer, J. M., Wang, S., Fischer, D. A., & Foreman-Mackey, D. 2018, *ApJ*, **867**, L3
- Cao, L., Pinsonneault, M. H., Hillenbrand, L. A., & Kuhn, M. A. 2022, *ApJ*, **924**, 84
- Capistrant, B. K., Soares-Furtado, M., Vanderburg, A., et al. 2024, *AJ*, **167**, 54

<sup>11</sup> <https://chancellorssciencescholars.unc.edu/>



- Chambers, K. C., Magnier, E. A., Metcalfe, N., et al. 2016, arXiv:1612.05560
- Cummings, J. D., Deliyannis, C. P., Maderak, R. M., & Steinhauer, A. 2017, *AJ*, **153**, 128
- Curtis, J. L., Agüeros, M. A., Matt, S. P., et al. 2020, *ApJ*, **904**, 140
- Dai, F., Goldberg, M., Batygin, K., et al. 2024, *AJ*, **168**, 239
- Damasso, M., Locci, D., Benatti, S., et al. 2024, *A&A*, **690**, A235
- Dang, L., Allart, R., Behounkova, M., et al. 2025, JWST Proposal ID. #8864, Cycle 4
- David, T. J., & Hillenbrand, L. A. 2015, *ApJ*, **804**, 146
- Dawson, R. I., Lee, E. J., & Chiang, E. 2016, *ApJ*, **822**, 54
- Feinstein, A., Welbanks, L., Ahrer, E.-M., et al. 2024, JWST Proposal ID. #5959, Cycle 3
- Feng, F., Butler, R. P., Vogt, S. S., et al. 2022, *ApJS*, **262**, 21
- Fernandes, R. B., Mulders, G. D., Pascucci, I., et al. 2022, *AJ*, **164**, 78
- Foreman-Mackey, D. 2018, *MNRAS*, **2**, 31
- Foreman-Mackey, D., Agol, E., Ambikasaran, S., & Angus, R. 2017, *AJ*, **154**, 220
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, *PASP*, **125**, 306
- Fritzewski, D. J., Barnes, S. A., James, D. J., & Strassmeier, K. G. 2021, *A&A*, **652**, A60
- Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, *A&A*, **674**, A1
- Galicher, R., Marois, C., Macintosh, B., et al. 2016, *A&A*, **594**, A63
- Giacalone, S., & Dressing, C. D., 2020 triceratops: Candidate Exoplanet Rating Tool, Astrophysics Source Code Library, ascl:2002.004
- Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, *AJ*, **161**, 24
- Gilbert, E. A., Barclay, T., Quintana, E. V., et al. 2022, *AJ*, **163**, 147
- Gossage, S., Conroy, C., Dotter, A., et al. 2019, *ApJ*, **887**, 199
- Hamer, J. H., & Schlafman, K. C. 2024, *AJ*, **167**, 55
- Hattori, S., Foreman-Mackey, D., Hogg, D. W., et al. 2022, *AJ*, **163**, 284
- Hedges, C., Hughes, A., Zhou, G., et al. 2021, *AJ*, **162**, 54
- Jeffries, R. D., Jackson, R. J., Franciosini, E., et al. 2017, *MNRAS*, **464**, 1456
- Jeffries, R. D., Jackson, R. J., Wright, N. J., et al. 2023, *MNRAS*, **523**, 802
- Jeffries, R. D., Littlefair, S. P., Naylor, T., & Mayne, N. J. 2011, *MNRAS*, **418**, 1948
- Jenkins, J. M., Twicken, J. D., McCauliff, S., et al. 2016, *Proc. SPIE*, **9913**, 99133E
- Johnson, M. C., Dai, F., Justesen, A. B., et al. 2018, *MNRAS*, **481**, 596
- Kipping, D. M. 2013, *MNRAS*, **435**, 2152
- Kirkpatrick, J. D., Cruz, K. L., Barman, T. S., et al. 2008, *ApJ*, **689**, 1295
- Kreidberg, L. 2015, *PASP*, **127**, 1161
- Lallement, R., Babusiaux, C., Vergely, J. L., et al. 2019, *A&A*, **625**, A135
- Lightcurve Collaboration, Cardoso, J. V. d. M., Hedges, C., et al., 2018  
Lightkurve: Kepler and TESS time series analysis in Python, Astrophysics Source Code Library, ascl:1812.013
- Lissauer, J. J., Ragozzine, D., Fabrycky, D. C., et al. 2011, *ApJS*, **197**, 8
- Lomb, N. R. 1976, *Ap&SS*, **39**, 447
- Lopez, E. D., & Fortney, J. J. 2013, *ApJ*, **776**, 2
- MacDougall, M. G., Petigura, E. A., Gilbert, G. J., et al. 2023, *AJ*, **166**, 33
- Mann, A. W., Gaidos, E., Mace, G. N., et al. 2016, *ApJ*, **818**, 46
- Mann, A. W., Vanderburg, A., Rizzuto, A. C., et al. 2018, *AJ*, **155**, 4
- Mann, A. W., Wood, M. L., Schmidt, S. P., et al. 2022, *AJ*, **163**, 156
- Meibom, S., Mathieu, R. D., Stassun, K. G., Liebesny, P., & Saar, S. H. 2011, *ApJ*, **733**, 115
- Ment, K., Fischer, D. A., Bakos, G., Howard, A. W., & Isaacson, H. 2018, *AJ*, **156**, 213
- Michel, K.-U., & Mugrauer, M. 2024, *MNRAS*, **527**, 3183
- Minkowski, R. L., & Abell, G. O. 1963, in Basic Astronomical Data: Stars and Stellar Systems, ed. K. A. Strand (Chicago, IL: Univ. Chicago Press), 481
- Moranta, L., Gagné, J., Couture, D., & Faherty, J. K. 2022, *ApJ*, **939**, 94
- Morris, R. L., Twicken, J. D., Smith, J. C., et al. 2020, in Kepler Data Processing Handbook: Photometric Analysis, Kepler Science Document KSCI-19081-003, id. 6, ed. J. M. Jenkins (Baltimore, MD: STScI), 6
- Muirhead, P. S., Mann, A. W., Vanderburg, A., et al. 2015, *ApJ*, **801**, 18
- Nardiello, D., Malavolta, L., Desidera, S., et al. 2022, *A&A*, **664**, A163
- Newton, E. R., Mann, A. W., Kraus, A. L., et al. 2021, *AJ*, **161**, 65
- Newton, E. R., Rampalli, R., Kraus, A. L., et al. 2022, *AJ*, **164**, 115
- Osborn, H. P., Bonfanti, A., Gandolfi, D., et al. 2022, *A&A*, **664**, A156
- Pepe, F., Mayor, M., Galland, F., et al. 2002, *A&A*, **388**, 632
- Petigura, E. A., Rogers, J. G., Isaacson, H., et al. 2022, *AJ*, **163**, 179
- Polanski, A. S., Lubin, J., Beard, C., et al. 2024, *ApJS*, **272**, 32
- Poon, S. T. S., Nelson, R. P., Jacobson, S. A., & Morbidelli, A. 2020, *MNRAS*, **491**, 5595
- Pourbaix, D., Tokovinin, A. A., Batten, A. H., et al. 2004, *A&A*, **424**, 727
- Pribulla, T., Kreiner, J. M., & Tremko, J. 2003, *CoSka*, **33**, 38
- Prša, A., Kochoska, A., Conroy, K. E., et al. 2022, *ApJS*, **258**, 16
- Ratzenböck, S., Großschedl, J. E., Möller, T., et al. 2023, *A&A*, **677**, A59
- Rebull, L. M., Stauffer, J. R., Bouvier, J., et al. 2016, *AJ*, **152**, 114
- Reid, I. N., Brewer, C., Brucato, R. J., et al. 1991, *PASP*, **103**, 661
- Rieke, G. H., Su, K. Y. L., Stansberry, J. A., et al. 2005, *ApJ*, **620**, 1010
- Riello, M., De Angeli, F., Evans, D. W., et al. 2021, *A&A*, **649**, A3
- Rizzuto, A. C., Mann, A. W., Vanderburg, A., Kraus, A. L., & Covey, K. R. 2017, *AJ*, **154**, 224
- Rowe, J. F., Bryson, S. T., Marcy, G. W., et al. 2014, *ApJ*, **784**, 45
- Scargle, J. D. 1982, *ApJ*, **263**, 835
- Smith, J. C., Stumpe, M. C., Van Cleve, J. E., et al. 2012, *PASP*, **124**, 1000
- Stefánsson, G., Kopparapu, R., Lin, A., et al. 2020, *AJ*, **160**, 259
- Stumpe, M. C., Smith, J. C., Catanzarite, J. H., et al. 2014, *PASP*, **126**, 100
- Stumpe, M. C., Smith, J. C., Van Cleve, J. E., et al. 2012, *PASP*, **124**, 985
- Thao, P. C., Mann, A. W., Barber, M. G., et al. 2024, *AJ*, **168**, 41
- Tian, H.-J., El-Badry, K., Rix, H.-W., & Gould, A. 2020, *ApJS*, **246**, 4
- Tofflemire, B. M., Rizzuto, A. C., Newton, E. R., et al. 2021, *AJ*, **161**, 171
- Tokovinin, A. 2018, *ApJS*, **235**, 6
- Twicken, J. D., Clarke, B. D., Bryson, S. T., et al. 2010, *Proc. SPIE*, **7740**, 774023
- Vach, S., Quinn, S. N., Vanderburg, A., et al. 2022, *AJ*, **164**, 71
- Vach, S., Zhou, G., Huang, C. X., et al. 2024, *AJ*, **167**, 210
- Valizadegan, H., Martinho, M. J. S., Jenkins, J. M., et al. 2023, *AJ*, **166**, 28
- Van Eylen, V., & Albrecht, S. 2015, *ApJ*, **808**, 126
- van Saders, J. L., Ceillier, T., Metcalfe, T. S., et al. 2016, *Natur*, **529**, 181
- Vanderburg, A., Huang, C. X., Rodriguez, J. E., et al. 2019, *ApJL*, **881**, L19
- Weiner Mansfield, M., Bean, J. L., Brady, M., et al. 2024, JWST Proposal ID. #4818, Cycle 3
- Weiss, L. M., Marcy, G. W., Petigura, E. A., et al. 2018, *AJ*, **155**, 48
- Weisserman, D., Becker, J. C., & Vanderburg, A. 2023, *AJ*, **165**, 89
- Wittrock, J. M., Dreizler, S., Reefe, M. A., et al. 2022, *AJ*, **164**, 27
- Wood, M. L., Mann, A. W., Barber, M. G., et al. 2023a, *AJ*, **165**, 85
- Wood, M. L., Mann, A. W., Barber, M. G., et al. 2023b, *AJ*, **166**, 247
- Wood, M. L., Mann, A. W., & Kraus, A. L. 2021, *AJ*, **162**, 128